

Q1: The n^{th} partial sum, S_n , of the series $\sum_{k=1}^{\infty} 5$ is equal to:

a) ∞

b) $5n$

c) n

d) 0

Q2: The sum of the series $\sum_{k=3}^{\infty} \frac{1}{2^k}$ is

a) $\frac{1}{8}$

b) $\frac{1}{4}$

c) $\frac{1}{2}$

d) $\frac{1}{16}$

Q3: The sequence $\sum_{n=0}^{\infty} (-1)^n + 5$ $\sum_{n=0}^{\infty} = 1$

a) converge to 5

b) converge to 6

c) diverges

d) converges to 4



Q 4: Applying the root test on the series $\sum \left(1 - \frac{1}{k}\right)^{k^2}$, then the value of $p = ??$

a) e

b) 1

c) e^{-1}

d) ∞

Q 5 :- IF the n^{th} partial sum of the series $\sum_{k=1}^{\infty} a_k$ is $S_n = -\tan^{-1}(n+1)$ then:-

a) the sum of this series is (0)

b) the sum of this series is $\left(\frac{\pi}{2}\right)$

c) the sum of this series is $\left(-\frac{\pi}{2}\right)$

d) this series is divergent.

Q 6: The sequence $\{1, 1, 0, 0, 1, 1, 2, 4, 6, 8, 10, \dots\}$

a) increasing

b) eventually strictly increasing.

c) eventually strictly decreasing.

d) strictly increasing.



Q 7: Find the interval of convergence of the power series $\sum_{k=1}^{\infty} \left(\frac{(-1)^k x^k}{k} \right)$?

a) $[-1, 1]$

b) $[-1, 1)$

c) $(-1, 1]$

d) $(-1, 1)$

Q 8: Which of the following series is convergent

a) $\sum_{k=1}^{\infty} \frac{\ln k}{k}$

b) $\sum_{k=1}^{\infty} \left(1 + \frac{1}{k} \right)^k$

c) $\sum_{k=1}^{\infty} \frac{1}{\sqrt{k}}$

d) $\sum_{k=1}^{\infty} \frac{1}{k!}$

Q 9: The sum of the series $\sum_{k=3}^{\infty} \frac{1}{2^k}$ is

a) $\frac{1}{16}$

b) $\frac{1}{4}$

c) $\frac{1}{8}$

d) $\frac{1}{2}$



Q 10: The sequence $\left\{ \frac{6n^2+1}{n^3+n+5} \right\}_{n=1}^{\infty} = 1$:

- a) converges to (2)
- b) converges to (0)
- c) converges to (6)
- d) diverges.

Q 11 : which the following series is an ~~alternating~~ alternating series :-

a) $\sum_{k=1}^{\infty} \frac{(-1)^k \cos(\pi k)}{k}$

b) $\sum_{k=1}^{\infty} \frac{\cos k}{\sqrt{k}}$

c) $\sum_{k=1}^{\infty} \frac{\sin k}{k}$

d) $\sum_{k=1}^{\infty} \frac{\cos(\pi k)}{\sqrt{k}}$

